

## **PROGRAM 8**

**Create kubernetes pods using YAML description**

**Project structure**

**1.nodejs\_configmap.yaml**

**2.nodejs\_pod.yaml**

**[3.server.js](#)**

**Step1:**

**Minikebe start**

**Step2:**

**Minikube status**

**Step 3: create these files**

```
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530: ~/Docke... x 1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530: ~/Docke... x 1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530: ~/Docke... x
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program-8$ cat nodejs-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });
1rv24mc010_ajayml@ajay-m-l-Inspiron-15-3530:~/Dockerfiles/program-8$ cat nodejs-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: program-8
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
      ports:
        - containerPort: 3000
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

## Step 4: Apply the config file and pod

```
1rv24mc010_ajayml@ajaym-l-Inspiron-15-3530: ~/Dockerfiles/program-8$ ls
nodejs-configmap.yaml nodejs-pod.yaml server.js
1rv24mc010_ajayml@ajaym-l-Inspiron-15-3530:~/Dockerfiles/program-8$ kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
mycron-29435320-jhncm               0/1     Completed 0           3m2s
mycron-29435321-gvznw               0/1     Completed 0           2m2s
mycron-29435322-l24nb               0/1     Completed 0           62s
mycron-29435323-gvr99               0/1     ContainerCreating 0           2s
mydeploy-8d4c44bfc-k4v94            1/1     Running   7 (4h15m ago) 17d
myjob-s6hdk                         0/1     Completed 0           17d
nginx-deployment-6f9664446b-6clkz   1/1     Running   2 (4h15m ago) 7d
nginx-deployment-6f9664446b-fxgbx   1/1     Running   2 (4h15m ago) 7d
nodejs-app                          0/1     CrashLoopBackOff 19 (2m50s ago) 6d23h
1rv24mc010_ajayml@ajaym-l-Inspiron-15-3530:~/Dockerfiles/program-8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config created
1rv24mc010_ajayml@ajaym-l-Inspiron-15-3530:~/Dockerfiles/program-8$ kubectl apply -f nodejs-pod.yaml
pod/program-8 created
1rv24mc010_ajayml@ajaym-l-Inspiron-15-3530:~/Dockerfiles/program-8$ kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
mycron-29435320-jhncm               0/1     Completed 0           3m32s
mycron-29435321-gvznw               0/1     Completed 0           2m32s
mycron-29435322-l24nb               0/1     Completed 0           92s
mycron-29435323-gvr99               0/1     ImagePullBackOff 0           32s
mydeploy-8d4c44bfc-k4v94            1/1     Running   7 (4h15m ago) 17d
myjob-s6hdk                         0/1     Completed 0           17d
nginx-deployment-6f9664446b-6clkz   1/1     Running   2 (4h15m ago) 7d
nginx-deployment-6f9664446b-fxgbx   1/1     Running   2 (4h15m ago) 7d
nodejs-app                          0/1     CrashLoopBackOff 19 (3m20s ago) 6d23h
program-8                           1/1     Running   0           8s
1rv24mc010_ajayml@ajaym-l-Inspiron-15-3530:~/Dockerfiles/program-8$ kubectl port-forward pod/program-8 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
```

# Output:

